

Master Thesis

Evaluation von Embeddings aus Foundation Modellen zur
Ausreißererkennung aus multivariaten tabellarischen Daten.

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Evaluating Foundation Model Embeddings for Anomaly Detection in Multivariate Tabular Data

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Abstract—Tabular data is the most common data format in practice, and real world tables combine numeric attributes with free text columns that are usually discarded during preprocessing. Recent work has shown that embedding these columns improves predictive performance for Tabular Foundation Models (TFMs), but that evidence is restricted to supervised classification and regression. This work asks whether the same holds for unsupervised anomaly detection, where no label guides the model towards the relevant part of the representation. Two datasets that differ in their share of free text are used, and an attribution analysis first establishes that the free text columns carry information relevant to the prediction. Across three TFMs, hyperparameter optimised baseline detectors and two families of text representations, no configuration achieves a consistent advantage. A FastText representation improves the detection on the text rich dataset, but the benefit does not transfer to the more numeric one, so in contrast to the supervised case no representation strategy makes the free text usable for unsupervised detection in general. A supervised control shows that the same models and representations perform well once labels are available. The obstacle therefore lies not in the quality of the learned representations but in their transfer into an unsupervised detection procedure.

Index Terms—tabular foundation models, transformer, anomaly detection, outlier detection, representation learning, embeddings, tabular data

I. INTRODUCTION

The number of scientific publications on Foundation Models (FMs) has grown rapidly in recent years [1]. An FM is a Machine Learning (ML) model that is trained on large amounts of data and can afterwards be applied to a wide range of tasks [2]. Such models perform transfer learning, which means that they apply knowledge acquired during pretraining to unseen tasks without further training [3]. Within current research, however, the studied modalities are distributed unevenly. Most publications address language and image generation, whereas FMs for tabular data and time series account for only a small share of the literature [1].

This imbalance is difficult to reconcile with the practical relevance of the tabular modality. Heterogeneous tabular data is the most frequently used data format in practice [4], and ML models are applied to it across critical domains such as cyber security [5], healthcare [6] and finance [7]. One task that is particularly relevant in these domains is anomaly detection, which finds data points that do not fit the pattern of the majority of the dataset [8]. Typical applications include the diagnosis of rare diseases, the detection of intrusions

in information technology systems and the identification of fraudulent financial transactions [8].

Tables in practice are multivariate and combine numeric attributes with free text columns, that is unstructured natural language fields such as job descriptions, clinical notes or product summaries. These columns are usually removed during preprocessing, which discards the semantic information they carry. How this information should instead be represented so that a TFM can exploit it has recently been studied by Mráz et al. [9]. They curate a benchmark of tabular datasets with meaningful text columns, and report that embedding these columns improves predictive performance, without a single embedding procedure being consistently superior. Two properties of that study frame the present work. It is restricted to supervised classification and regression, and it considers only one family of representations, namely external embeddings that encode a text cell in isolation.

The first restriction defines the gap addressed here. Under supervision a label guides the model towards the part of the representation that is relevant for the target. In unsupervised anomaly detection no such guidance exists, and the detector must rely on the structure of the input distribution alone. It is therefore not evident that the benefit of text embeddings observed in the supervised case carries over [9]. Whether TFMs are useful for unsupervised anomaly detection at all, and whether the semantics of free text columns can be made accessible to a detector without a label, remains open.

This work investigates both questions on two datasets that differ in their share of free text. One of them is part of the benchmark of Mráz et al. [9], which allows the findings of the unsupervised setting to be placed directly against their supervised results. TFMs and established baseline detectors are compared on a common data basis, and the influence of different representation forms of the free text columns is isolated. The study is guided by three research questions.

- **RQ1.** How do TFMs perform in an unsupervised setting compared to established baseline detectors?
- **RQ2.** Which representation strategy, if any, makes the information of the free text columns usable for the baseline detectors?
- **RQ3.** Do the representations of the TFMs become effective once a label is available, so that a weak unsupervised result can be attributed to the transfer into the unsupervised procedure rather than to the quality of the representations?

RQ2 is deliberately not restricted to text embeddings. This addresses the second restriction of the study of Mráz et al [9]. If only embeddings of the text cells were compared, a negative result would leave open whether the hurdle lies in the specific embedding procedure or in the representation of the row as a whole. A model-internal representation of a TFM is therefore included as a second and structurally different source. It encodes the position of a row within the table rather than the meaning of a text cell. Because it is obtained by fitting a TFM on the label, it carries label information and is therefore semi-supervised. The label enters the construction of the representation, although the detector itself receives no label at inference.

The three questions further presuppose that the free text columns carry information that is relevant for the detection at all. This assumption is not self evident, and a negative answer to RQ2 would be uninterpretable without it, since it would remain unclear whether the failure lies in the text or in its representation. The assumption is therefore verified in a preliminary analysis that precedes the three experiments.

The contributions of this work are the following.

- An attribution analysis showing that free text columns carry information that is relevant for the prediction and that they dominate the most influential attributes on the text-rich dataset. This establishes the premise for the three experiments (Section IV-A).
- A systematic comparison of three TFMs and hyperparameter optimised baseline detectors under an identical unsupervised protocol, including their runtime (Section IV-B).
- An ablation over two families of representations. The first covers external text embeddings and compares FastText, which performed best most often in the study of Mráz et al. [9], with a sentence transformer. The second covers model internal representations of a TFM, which are not considered in prior work. Neither family improves the unsupervised detection performance across both datasets, and the one variant that helps on the text-rich dataset does not transfer to the more numeric one, in contrast to the supervised case (Section IV-C).
- A supervised control experiment showing that the weak unsupervised performance of the TFMs does not originate from insufficient representation quality but from their adaptation to the unsupervised procedure (Section IV-D).

Taken together, these contributions locate the obstacle in the unsupervised transfer of the representations rather than in the free text or in the quality of the models. Section V develops this reading across the four sets of results.

The remainder of this paper is structured as follows. Section II introduces the relevant paradigms and positions this work within the state of the art. Section III describes the datasets, the representation variants, the models and the evaluation protocol. Section IV presents the preliminary analysis and the three experiments together with their results. Section V discusses the findings across the experiments and states the limitations of the study. Section VI concludes and outlines directions for future work. All source code and the raw data

are publicly available. The preprocessed data and the complete experiment logs are not published directly but are reproduced by running the released pipeline (see footnotes).^{1 2 3}

II. BACKGROUND AND RELATED WORK

A. Foundation Models for Tabular Data

FMs differ from baseline detectors in the origin of their knowledge. A baseline detector derives its notion of what is normal from the dataset at hand alone. An FM carries knowledge from a large pretraining corpus and applies it to a new table without being retrained on it [2]. Within the tabular modality three paradigms can be distinguished. They differ in the kind of information they exploit, and this difference is the axis along which the present work is organised.

a) *Prior-data Fitted Networks*: Prior-data Fitted Networks (PFNs) [10] are trained on synthetic data that is generated from predefined statistical priors [11]. A prior in the Bayesian sense is an initial assumption about an unknown parameter before any data has been observed [12]. A model that relies on such priors therefore bases its predictions on assumptions about distributions and frequencies, without considering the semantic meaning of column names or categorical values. Inference is performed through In-Context Learning (ICL), which allows the model to predict from a small set of context examples without being trained on the specific dataset. The context examples act as the observed data that turns the prior into a posterior, from which the prediction is drawn.

Formally, a PFN approximates the posterior predictive distribution over the target y for a query x given a context dataset D [10]:

$$p(y | x, D) \propto \int_{\Phi} p(y | x, \phi) p(D | \phi) p(\phi) d\phi, \quad (1)$$

where a hypothesis $\phi \in \Phi$ is a data generating mechanism and Φ is the space of these mechanisms, which the prior defines [11]. Each hypothesis is weighted by its prior probability $p(\phi)$ and by the likelihood $p(D | \phi)$ of the context under it. The integral is intractable, so the network is trained on synthetic datasets sampled from the prior to approximate this distribution directly with a single forward pass, which is what enables the label free inference used for anomaly detection in this work. The approximation is realised by a Transformer [13] in which each feature vector and its label form one token, and context and query tokens exchange information through self-attention in a single forward pass.

TabPFN [14] is the most prominent representative and outperforms shallow approaches on classification and regression without additional tuning. Alongside the supervised variant an unsupervised variant is provided that requires no label and can be applied to anomaly detection directly. FoMo-0D [15] differs in that it was not derived from a classification model but was designed for tabular outlier detection from the outset. Both

¹<https://github.com/florivz/tfm-anomaly-detection>

²<https://insideairbnb.com/get-the-data/>

³<https://www.kaggle.com/datasets/hassanlegends/fake-job-postings>

models process numeric input only and discard the semantics of free text.

b) LLM based Models: The second paradigm inverts this relation and includes a Large Language Model (LLM). AnoLLM [16] serialises each table row into a textual sequence and processes it with a pretrained language model. Free text therefore requires no prior vectorisation and its semantics remains available to the model. The cost is that numeric values have to pass through a textual representation, which can weaken their mathematical precision.

c) Hybrid Models: ConTextTab [17] combines the PFN architecture with sentence transformer embeddings. Textual and categorical cells are converted into vectors by a pretrained sentence embedding model, and the column names are encoded in the same way and act as a semantic positional encoding. Categorical values therefore retain their meaning instead of being replaced by a meaningless one-hot encoding. The model is trained on real tables rather than on synthetic priors. It is restricted to classification and regression and does not perform anomaly detection.

B. Baseline Detectors

Established baseline detectors from the benchmark paper [8] serve as the reference against which the TFMs are measured. They proved to be strong standards in the ADBench benchmark [8] and differ in how they formalise the notion of an outlier. Isolation Forest does not model what normal points look like but isolates anomalies through random binary trees, exploiting the observation that rare and distant points can be separated with few random splits [18]. ECOD [19] evaluates the tail probabilities of empirical cumulative distribution functions and requires no hyperparameters. LODA [20] aggregates one dimensional histograms over random projections. The autoencoder [21] uses the reconstruction error as the anomaly score, under the assumption that normal points are reconstructed more accurately than anomalous ones.

C. Embeddings and Dimensionality Reduction

An embedding maps an object into a dense vector space in such a way that similar objects lie close to each other. Unlike one-hot encoding, which represents each level as a separate column, or frequency encoding, which replaces a category by its frequency, an embedding preserves relations of similarity in the representation itself.

The embeddings in this work differ in their frame of reference. A sentence transformer encodes the meaning of a text field in isolation and answers the question of how similar two texts are in content. A TFM produces a context dependent representation that positions a data point within the structure of the entire table. The first captures semantics, the second captures tabular context.

Three sources of representations are used in this work, and they differ in what they encode. FastText builds a vector for a text cell by averaging its word vectors, which captures lexical content but discards word order and sentence context [22]. The

sentence transformer encodes the same cell in its sentence-level context and captures the meaning of the field rather than its words alone. Both remain confined to the content of a single cell, whereas the model-internal representation of a TFM describes the position of an entire row within the structure of the table. This distinction is what makes the comparison informative. If the detection depended only on the meaning of the text, the external embeddings should help, whereas if it depended on the position of a row in the tabular distribution, the model-internal representation should help. Experiment 2 tests which of the two, if either, transfers to an unsupervised detector.

Both produce a high-dimensional input space, and this creates a problem for the detectors. As the dimensionality grows, pairwise distances become increasingly uniform under the curse of dimensionality [23], which reduces the discriminative power of distance-based and density-based detectors. The common remedy is a Principal Component Analysis (PCA), which projects the data onto orthogonal axes of maximal variance and discards components with low variance. This reduction is not neutral. It structurally favours information that is redundantly encoded across many correlated dimensions, whereas single highly informative dimensions with a small share of variance can be lost. A supervised attribution method such as SHapley Additive exPlanations (SHAP) [24], which is the strongest feature selector in [9], is not available here, since no label is present. An unsupervised PCA is therefore used. This property is the theoretical basis for RQ2 and is examined empirically in Section IV-C.

D. Enhanced Baseline Models

This work introduces the term enhanced baseline models for shallow detectors, that is detectors without a deep neural architecture, that operate on representations produced by a TFM instead of on the raw attributes. The concept follows from a conflict that runs through the three paradigms above. Purely numeric models leave the information contained in semantic attributes largely unused, which can reduce detection performance on complex datasets. Purely semantic approaches carry the opposite risk, since numeric values lose mathematical precision when they are serialised into text or absorbed into embedding vectors. This tension between statistical exactness and semantic density is what the experiments in this paper make visible. The semantic density of a dataset denotes the share of its attributes that consist of free text, as opposed to numeric or categorical attributes. The two datasets in Section III-A were chosen to differ precisely in this property.

E. Closest Related Work and Research Gap

The work most closely related to this paper is the study of Mráz et al. [9]. It addresses the same question, namely how free text columns should be represented so that TFMs can exploit them, and it makes three observations that frame the present work.

First, the authors report that established tabular benchmarks almost never contain textual columns and that real datasets

with semantically rich text are hard to find, so that TFMs are rarely evaluated on tabular data with text at all. They therefore curate a benchmark under five rules. One requires a dual signal, so that a dataset counts only if predictive value lies in the text as well as in the structured columns, which rules out cases that are effectively text-only or table-only. The Fake Job Postings dataset used in this paper is part of that benchmark, which places the present results on a directly comparable data basis. Second, they evaluate several strategies for embedding a text column into the numeric input space and find that each fails under a different condition. Bag of words procedures break down under unseen synonyms, FastText loses the target signal under random noise in long texts, and language model embeddings are impaired by semantic ambiguity. No single procedure is consistently superior, although FastText performs best most often. This motivates the use of two different embedding procedures in this paper rather than a single one, with FastText serving as the point of contact with their results. Third, they report that such embeddings improve performance over the text free variant and conclude that text in tabular models remains an unsolved problem. This finding holds for supervised classification and regression, which is the only setting they consider.

Two gaps follow from this. The first concerns the unsupervised learning setting. In unsupervised anomaly detection this guidance is absent, and the detector has to identify deviations from the structure of the input distribution alone. It is therefore open whether the reported benefit of text embeddings carries over. The second concerns the space of representations under study. All procedures compared by Mráz et al. [9] belong to a single family. They encode the content of a text cell in isolation and append the resulting vector to the table. A negative result within this family alone would remain ambiguous, since it would not distinguish between a weakness of the individual embedding procedure and a weakness of the underlying idea of representing the text separately from the row. The present work therefore adds a second and structurally different family, namely the model-internal representations of a TFM introduced in Section II-D. These encode the position of a row within the table rather than the meaning of a text cell, and they are not considered in prior work on tabular data with text.

The present work addresses both gaps. It transfers the question to unsupervised anomaly detection and it compares the two families of representations on a common data basis. It further separates the question of whether the free text carries information at all from the question of whether a given representation makes that information accessible to a detector. The dual signal requirement of Mráz et al. [9] is verified for the anomaly detection task in the preliminary analysis in Section IV-A.

III. METHODOLOGY

A. Datasets

Two datasets are used. They are selected so that they cover a spectrum between a mostly numeric table and a table with

TABLE I
PROPERTIES OF THE TWO DATASETS AFTER CLEANING. THE COUNTS REFER TO SOURCE COLUMNS, NOT THE ENCODED FEATURE MATRIX (WHICH HAS MORE COLUMNS). THE FREE TEXT COLUMNS ARE NOT PART OF THE VARIANT `CLEANED` AND ENTER ONLY THE VARIANT `CLEANED_TEXT`.

	Fake Job Postings	Airbnb Paris
Rows	17,880	18,350
Anomalies	866 (4.84 %)	721 (3.93 %)
Numeric and binary	5	26
Categorical after encoding	6	11
Geographic	3	2
Sum in <code>cleaned</code>	14	39
Free text	5	4
Attributes in total	19	43
Label	native	constructed

a high share of free text. This contrast is required in order to test whether the benefit of textual representations depends on the semantic density of the data. The Fake Job Postings dataset is part of the benchmark curated by Mráz et al. [9] and therefore satisfies their dual signal requirement, which states that both the textual and the structural attributes must carry predictive information. This makes the results of this paper directly comparable to their supervised findings. Table I summarises the properties of both datasets. The counts refer to the source columns and not to the columns of the encoded feature matrix, which is larger. For the Airbnb dataset the reported figures refer to the state after the construction of the anomaly label, in which the middle rating segment has been removed.

The Real or Fake Job Posting Prediction dataset [25], obtained from the public re-upload used in this work [26], contains job advertisements and provides a native anomaly label. The share of fraudulent postings is low, which makes the dataset a standard anomaly detection problem. The dataset contains a high number of missing values, for example in `salary_range` and `department`. Throughout this paper a dataset is called text-rich if a comparatively large share of its attributes are free text columns, which holds for Fake Job Postings and less so for Airbnb Paris.

The *Airbnb Paris* dataset is based on data from the Inside Airbnb platform [27] and covers listings scraped in September 2025. The restriction to a single city ensures that all entries share the same language and the same column structure, which is a precondition for the generation of the text embeddings and for the use of an English sentence embedding model. The dataset has no native anomaly label. The label is therefore constructed from the user ratings. Listings with a rating of 5.0 are treated as inliers and listings with a rating of at most 3.0 as outliers, while the middle rating segment is removed in order to obtain a clear separation of the two classes. The implications of this construction are addressed in Section V-A.

B. Data Preparation and Representation Variants

All experiments operate on a common preprocessing basis. The cleaned pipeline produces a purely numeric table in which the free text columns are removed, categorical attributes are encoded and numeric attributes are scaled. This variant is required because the baseline detectors as well as the table-native TFMs, that is models that consume the table column wise without serialising a row into a text sequence, cannot process text natively. Categorical attributes with few levels are one-hot encoded, whereas attributes with many levels are frequency encoded in order to avoid an unnecessarily high-dimensional and sparse input space. Missing values in numeric attributes are imputed with the median and missing values in boolean attributes with the mode. Categorical attributes receive an explicit sentinel category instead, so that a missing value remains visible to the models rather than being merged into the most frequent level. Columns that are constant are dropped. For Fake Job Postings an indicator column records whether the salary range is missing, since the absence of this field is itself informative. For Airbnb Paris all review subscores and all review counters are removed, since they enter the construction of the label and would otherwise leak it. This includes the number of reviews of the last twelve months, which is the last remaining review counter and a direct proxy of the label. The tenure of a host is computed against the fixed reference date of the scrape rather than against the latest date in the data, so that the value of a row does not depend on which rows are present.

All transformations are fitted on the training rows alone. This applies to the median and mode imputation, the frequency encoding, the removal of constant columns, the standard scaling, the PCA reduction, the FastText model and the TabPFN embeddings. The validation rows and the test rows are only transformed with the objects that were fitted on the training rows. A category that does not occur in the training rows receives a frequency of zero. The QuantileTransformer of FoMo-0D is fitted on the context subsample of 5000 rows, which is drawn from the training rows as well. TabPFN-Unsupervised draws its context in the same way, capped at 3000 rows by the available memory. Both contexts are drawn at random and therefore contain the anomalies at their natural rate. The sentence transformer is an exception in that it is used as a frozen encoder and is not fitted at all, so that its embeddings do not depend on the split, whereas the PCA that reduces them is again fitted on the training rows. The FastText model is trained with a single thread, because the parallel mode of the implementation is not deterministic. Its internal random number generator is fixed by the binding and cannot be set, so the variation across the seeds originates from the different training corpus of each split. The enhanced representations form a separate case that is treated in Section II-D and Section V-A, since they are fitted on the label and are therefore semi-supervised.

On top of the `cleaned` table a set of variants is constructed that differ only in how the free text columns are treated. These

variants form the object of study of RQ2. Table II lists them. All variants share the identical numeric base of the cleaned table, so that a difference between them can be attributed to the treatment of the free text alone.

The variant `cleaned_text` appends the original free text columns and serves as the input for the models that process text natively, namely AnoLLM and ConTextTab. The variants `semantic` and `fast_text` replace each free text column by an embedding vector. The embedding is computed per column, and the embedding of the column name is added to the cell embedding so that the assignment of a text to its column is encoded explicitly. Both variants differ only in the underlying embedding model, which makes it possible to test whether the choice of the embedding procedure affects the usability of the text for the detectors. Since the resulting input space is high-dimensional and thereby affects distance-based and density-based detectors, both variants are additionally provided with a PCA reduction to 100 and to 30 components. Both levels are evaluated rather than fixed in advance, because the share of variance that a given number of components retains differs by dataset and by embedding procedure, as Table IX in the appendix shows. The reduction to 30 components in particular discards a large part of the variance of the embedding space. Whether this loss harms the detection is an empirical question that Section IV-C answers.

The variant `enhanced` follows a different route and uses the internal representations of a TFM. TabPFN is fitted on the label of the text free `cleaned` table, and the resulting embeddings replace the original attributes. The detectors then operate on the model-internal representation instead of the raw attributes. This produces the enhanced baseline models introduced in Section II-D. Because TabPFN is fitted on the label, these variants carry label information. They are therefore classified as semi-supervised and are not directly comparable with the unsupervised variants. They are reported as a diagnostic upper bound rather than as a competing method.

The variant `enhanced_semantic_pca30` finally combines both representation sources through an inner join on the row identifier. It makes it possible to test whether the two sources complement each other or whether they contribute redundant information.

C. Models

The selection of the models follows two criteria, namely the reported performance on established benchmarks and the suitability for the experimental setup of this work. Table III gives an overview.

Among the TFMs, TabPFN [14] takes a special position, since it is the only model that is used both as an unsupervised detector, as a source of representations for the enhanced baselines and as a supervised classifier. FoMo-0D [15] is the only model that was developed specifically for tabular anomaly detection and therefore serves as the direct point of comparison for RQ1. AnoLLM [16] serialises each table row into a textual sequence and processes it with a language model, which makes it the counterpart of the table-native models. ConTextTab [17]

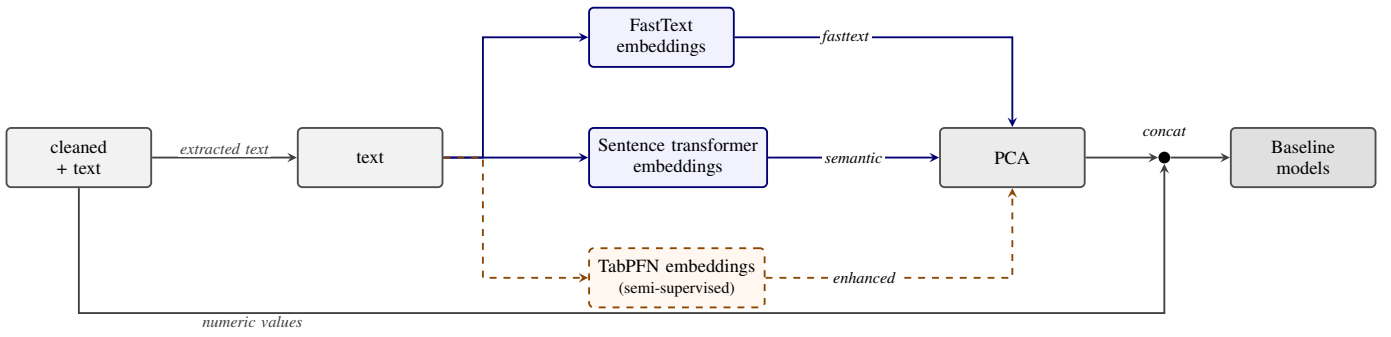


Fig. 1. Representation pipeline studied in Experiment 2. The free text columns are extracted from the cleaned table and encoded by one of three procedures. The two external text embeddings, FastText and the sentence transformer, are unsupervised. The TabPFN representation, shown with a dashed border, is obtained by fitting the model on the label and is therefore semi-supervised. Each embedding is reduced by a PCA and concatenated with the numeric attributes before it is passed to the baseline detectors. The PCA step is optional, and both the reduced and the unreduced variants are evaluated.

TABLE II
REPRESENTATION VARIANTS STUDIED IN THIS WORK. THE ENHANCED VARIANTS CARRY LABEL INFORMATION AND ARE THEREFORE SEMI-SUPERVISED. THE VARIANT `CLEANED_TEXT` SERVES THE MODELS THAT PROCESS TEXT NATIVELY AND IS NOT PART OF THE DETECTOR COMPARISON IN EXPERIMENT 2.

Variant	Text repr.	Setting
<code>cleaned</code>	removed	unsup.
<code>cleaned_text</code>	raw text	unsup.
<code>semantic_pca100</code>	sentence emb.	unsup.
<code>semantic_pca30</code>	sentence emb.	unsup.
<code>fast_text_pca100</code>	FastText	unsup.
<code>fast_text_pca30</code>	FastText	unsup.
<code>enhanced</code>	TabPFN	semi sup.
<code>enhanced_pca30</code>	TabPFN	semi sup.
<code>enhanced_semantic_pca30</code>	TabPFN + sent.	semi sup.

is restricted to classification and regression and is not used as a detector. It processes text columns natively and therefore treats each free text column as an individual attribute, which is the property that the attribution analysis in Section IV-A relies on.

The baseline detectors were chosen so that each covers a different anomaly paradigm and so that all of them remain applicable in high-dimensional spaces. They were required to scale to high dimensions and to remain independent of distance and density metrics, since the latent space of the embeddings stays high-dimensional and sparsely populated even after a PCA, where such measures lose their discriminative power under the curse of dimensionality. k Nearest Neighbors (kNN) and the Local Outlier Factor did not meet these requirements and were therefore excluded, as preliminary tests confirmed.

Two embedding models are used to vectorise the free text columns. The sentence transformer `all-mpnet-base-v2` [28], [29] produces 768 dimensional embeddings and follows the approach of ConTextTab, which also converts textual cells into vectors by means of a sentence embedding model. FastText [22] is added as a simpler and computationally cheaper alternative that averages word vectors and therefore ignores the sentence context. It is included because it achieved the

TABLE III
MODELS AND THEIR ROLE IN THE EXPERIMENTS.

Model	Paradigm	Role
TabPFN	PFN	detector, embedding extract., classifier
FoMo-0D	PFN	detector
AnoLLM	LLM	detector
ConTextTab	hybrid	classifier, attribution
Isolation Forest	classical	baseline detector
ECOD	classical	baseline detector
LODA	classical	baseline detector
Autoencoder	neural	baseline detector
<code>all-mpnet-base-v2</code>	sentence emb.	text vectorisation
FastText	word emb.	text vectorisation

best performance among the tested embedding procedures in a recent supervised study [9]. Neither model acts as a detector.

A more detailed comparison of the models, including their capacity limits and their role in this work, is given in Table X in the appendix.

D. Evaluation Protocol

Each dataset is split in a stratified manner into 50 % training data, 20 % validation data and 30 % test data. The split is drawn once per seed and stored as a file, so that every notebook of the pipeline operates on the same partition. This yields training, validation and test parts of comparable size on both datasets. All preprocessing steps and all models are fitted on the training part alone, as described in Section III-B. The validation part is used only for the selection of hyperparameters. The test part enters no fit call and is evaluated exactly once per run, after the selection is complete.

The detectors are fitted on the unlabeled training part in its original distribution, which contains the anomalies at their natural rate of 4.84 % and 3.93 % respectively. The setting is therefore unsupervised outlier detection on a contaminated training set and not one class novelty detection on clean data.

All models are evaluated with Area Under the Precision Recall Curve (AUPRC) and Area Under the Receiver Operating Characteristic Curve (AUROC) in order to keep the results

comparable with common benchmarks such as ADBench [8]. AUROC measures how well a model ranks anomalies above normal points, whereas AUPRC is the more demanding measure under the strong class imbalance of both datasets, since it reflects how well the rare class is identified. AUPRC is therefore treated as the primary metric throughout this work. All rankings and all model selections are made with respect to it, and AUROC is reported alongside as a secondary measure that is less sensitive to the class imbalance. The runtime is reported in addition, because a productive application has to take the resource demand into account. A classification report is produced for every run as an additional diagnostic during development, but it is not part of the reported results.

The baseline detectors are implemented with the PyOD library [30] and are tuned with an exhaustive grid search. The predefined search space is documented in Table XI. The search is repeated for every combination of dataset, experiment, representation, detector and seed, so that no configuration is carried over from one experiment or one representation to another. The candidate configurations are fitted on the training part, and the configuration with the highest AUPRC on the validation part is selected. The winning configuration is then refitted on the training part and evaluated once on the test part. The detectors themselves are unsupervised, since the fit uses unlabeled features only, but the selection of the configuration uses the labels of the validation part. In a strictly unsupervised scenario this information would not be available, so the baseline detectors are favoured over the untuned TFMs. This asymmetry is deliberate and works against the hypothesis of this work, since it strengthens the baselines rather than the TFMs. The TFMs are evaluated in their default configurations.

The grid search is not intended to tune the baseline detectors to peak performance but to prevent an unfavourable default configuration from degrading their results by chance. The selected configurations support this reading, since they are not stable across the seeds. In Experiment 1 no combination of dataset and detector selects the same configuration in all three seeds, and across both experiments only a small minority does. No grid point wins clearly more often than the alternatives in its search space. The optimum is therefore flat, and the spread of the results across the seeds originates from the split rather than from the choice of hyperparameters. For the same reason no table of selected configurations is reported, since a modal winner would suggest a stability that does not exist. The selected configuration of every run is logged and remains available. The consequences of this choice are discussed in Section V-A.

All experiments are repeated for the seeds 1, 2 and 3. The seed controls the split as well as the model initialisation and is logged with every run. The reported values are the mean over the three seeds, and the standard deviation is reported alongside them. The repetitions are therefore independent splits and not re-executions of a single pipeline.

The experiments were executed on Graphics Processing Unit (GPU) instances provided by inovex GmbH. The available memory limits the context size of TabPFN and the batch

size of AnoLLM, and it prevents the evaluation of the semantic and FastText variants without a preceding PCA reduction. These constraints are stated in Section V-A. All FMs are integrated through their official open source repositories and the released model checkpoints.

The AUPRC of a random ranking equals the share of anomalies in the test part, as reported in Table I. All AUPRC values in this work have to be read against that reference, which lies below five per cent on both datasets.

IV. EXPERIMENTS AND RESULTS

This section presents a preliminary analysis followed by three experiments. The preliminary analysis establishes that the free text columns carry information that is relevant for the prediction. Experiment 1 compares the TFMs with the baseline detectors in the unsupervised setting. Experiment 2 studies the representation variants. Experiment 3 uses supervised classification as a control.

A. Preliminary Analysis of the Information Content of the Free Text Columns

The three research questions presuppose that the free text columns carry information relevant to the detection. If they did not, a negative result for RQ2 would be uninterpretable, since the failure could not be attributed either to the text or to its representation. This analysis tests the premise directly. It is not an experiment on the detection task itself but a check that the dual signal requirement of Mráz et al. [9] holds for the two datasets.

The analysis uses ConTextTab, which processes text columns natively and therefore treats each free text column as a single attribute rather than as a bag of embedding dimensions. The model is fitted on an in-context training set of 600 rows that is balanced at a ratio of 1:4, with 120 outliers and 480 inliers drawn from the training part. Under the natural anomaly rate a context of this size would contain only about 25 outliers, which is not enough for the model to reach the predictive quality that a SHAP attribution presupposes. Before the attribution is computed, the model has to reach a sufficient predictive quality, since SHAP values are interpretable only for a model that predicts well. The gate requires an AUROC above 0.8 on the full test part in its natural distribution, which ConTextTab passes on both datasets in this configuration. The values in Section IV-D are obtained with a larger in-context set and are therefore reported separately. The attribution itself is computed with the model agnostic KernelExplainer of the SHAP library, since ConTextTab offers no differentiable access, and it explains the outlier probability that the model predicts. Each free text column enters as one feature, so that a perturbation replaces the entire text of a cell by the text of a background row. The explainer draws 100 samples per explained row. The background set consists of ten rows drawn at random from the in-context training set and therefore stems from the training part. The 30 explained rows, namely 10 outliers and 20 inliers, are drawn from the training part as well. The mean absolute SHAP value of each attribute

over the explained rows serves as a measure of its influence on the prediction, and its mean and standard deviation over the three seeds are reported. Bootstrap confidence intervals and measures of rank stability are not computed, which is noted as a limitation in Section V-A.

Table IV reports the ten most influential attributes. On the text-rich Fake Job Postings dataset the free text columns dominate. All five free text columns of the dataset appear among the ten most influential attributes, at ranks one, three, four, five and seven, and `company_profile` leads the table with a mean absolute SHAP value of 0.166, which exceeds the strongest non-textual attribute at 0.041 by about a factor of four. On the more numeric Airbnb Paris dataset the picture is more balanced. Two of the four free text columns appear among the ten most influential attributes, at ranks three and four, and their values of 0.026 and 0.025 lie close to the two leading attributes at 0.028. The free text is therefore not dominant on this dataset, but it ranks among the most influential attributes. The share of free text among the attributes of this dataset is small, as Table I shows, so the free text columns are clearly overrepresented among the ten most influential attributes relative to that share. Figure 2 aggregates the same attribution by attribute type and summarises this contrast. On Fake Job Postings the summed influence of the free text columns clearly exceeds that of the numeric attributes, whereas on Airbnb Paris the numeric attributes carry the larger summed influence and the free text contributes a smaller but non-negligible share. Since the two datasets differ in the number of columns of each type, the comparison should be read together with the per-attribute view of Table IV, which shows that the free text columns rank among the most influential attributes on both datasets. This pattern matches the different shares of free text in the two datasets. The premise for the three experiments therefore holds on both datasets, more clearly so where the share of text is higher, with the qualification that the analysis rests on a classification model and therefore speaks to the information content of the free text rather than to the behaviour of the detectors, as Section V-A notes.

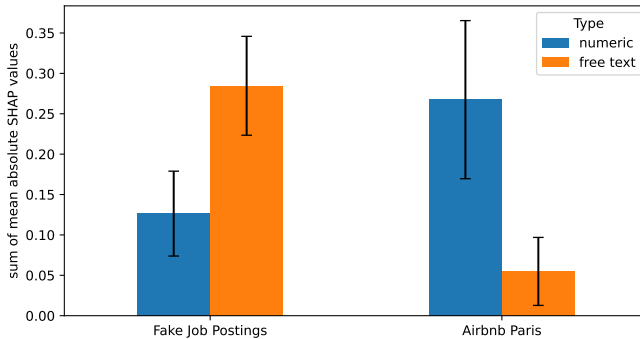


Fig. 2. Summed mean absolute SHAP value of the numeric and the free text attributes on both datasets. The free text carries the larger share on Fake Job Postings and a smaller share on Airbnb Paris, in line with the different shares of free text in the two datasets.

TABLE IV
PRELIMINARY ANALYSIS. THE TEN MOST INFLUENTIAL ATTRIBUTES BY MEAN ABSOLUTE SHAP VALUE ON THE FITTED CONTEXTTAB MODEL, AVERAGED OVER THREE SEEDS. TEXT ATTRIBUTES ARE SET IN BOLD.

Fake Job Postings		Airbnb Paris	
Attributes	SHAP	Attributes	SHAP
company_profile	0.166	host_listings_cnt_homes	0.028
has_company_logo	0.041	host_listings_count	0.028
description	0.038	name	0.026
title	0.034	description	0.025
requirements	0.029	amenities_count	0.019
salary_avg	0.019	est. occupancy 1365d	0.017
benefits	0.017	host_tenure_days	0.014
industry	0.010	instant_bookable	0.012
has_questions	0.010	availability_365	0.012
employment_type	0.009	availability_eoy	0.011

B. Experiment 1: Unsupervised Anomaly Detection Benchmark

Experiment 1 addresses RQ1. It compares the three TFMs that act as detectors, namely TabPFN-Unsupervised, FoMo-0D and AnoLLM, with the four baseline detectors on the text free `cleaned` variant. AnoLLM is the exception, since it serialises each row itself and therefore receives the readable raw values including the free text rather than the scaled numeric table. The baseline detectors are tuned by grid search as described in Section III-D.

Table V reports AUPRC, AUROC and runtime for both datasets. On Fake Job Postings AnoLLM reaches the highest point estimate of all models, although the difference to the tuned baseline detectors stays within one standard deviation. Its value is also not directly comparable, since AnoLLM is the only model in this experiment that receives the free text columns. It is followed closely by the tuned baseline detectors, and the two table native TFMs fall behind three of the four baseline detectors. On Airbnb the order changes and the two best models are the statistical detectors ECOD and Isolation Forest, ahead of every TFM. The weakest model on the Airbnb dataset is FoMo-0D, which reaches the lowest AUPRC of all models at 0.047 and an AUROC of 0.587, despite being the only model designed specifically for tabular anomaly detection. This result is unexpected and its cause is not established here. Since the model was pretrained for this task, the weak performance may point to a limited transfer of its priors to the two datasets studied rather than to a deficiency of the approach itself. The value of TabPFN-Unsupervised on Fake Job Postings has to be read with the constraint stated in Section V-A, since the context available to the model is capped. The evaluation itself covers the full test part, so the restriction concerns the information the model can draw on and not the basis of the measurement. Across both datasets no TFM achieves a consistent advantage over the tuned baseline detectors.

The runtime column reveals a second gap. The statistical detectors complete in well under three seconds, whereas AnoLLM requires roughly one and a half hours on both

TABLE V
EXPERIMENT 1. UNSUPERVISED ANOMALY DETECTION. AUPRC, AUROC (WITH STANDARD DEVIATIONS OVER THREE SEEDS) AND RUNTIME IN SECONDS. MODELS IN ITALICS ARE TFMS. BEST AUPRC PER DATASET IN BOLD.

Data	Model	AUPRC $\pm$ SD	AUROC $\pm$ SD	Time
Fake Jobs [26]	<i>AnoLLM</i>	0.092 $\pm$ 0.009	0.626 $\pm$ 0.011	5400.6
	Isolation Forest	0.084 $\pm$ 0.010	0.619 $\pm$ 0.015	2.3
	ECOD	0.076 $\pm$ 0.004	0.626 $\pm$ 0.009	0.6
	Autoencoder	0.074 $\pm$ 0.002	0.615 $\pm$ 0.011	143.3
	<i>FoMo-OD</i>	0.067 $\pm$ 0.006	0.635 $\pm$ 0.019	0.2
	<i>TabPFN-Unsup</i>	0.062 $\pm$ 0.003	0.524 $\pm$ 0.034	275.4
	LODA	0.055 $\pm$ 0.016	0.504 $\pm$ 0.069	0.5
Airbnb [27]	ECOD	0.100 $\pm$ 0.011	0.741 $\pm$ 0.014	0.8
	Isolation Forest	0.094 $\pm$ 0.019	0.721 $\pm$ 0.013	2.4
	LODA	0.084 $\pm$ 0.021	0.655 $\pm$ 0.021	0.6
	<i>TabPFN-Unsup</i>	0.075 $\pm$ 0.003	0.684 $\pm$ 0.007	954.1
	<i>AnoLLM</i>	0.072 $\pm$ 0.009	0.659 $\pm$ 0.012	5923.7
	Autoencoder	0.068 $\pm$ 0.009	0.645 $\pm$ 0.017	155.0
	<i>FoMo-OD</i>	0.047 $\pm$ 0.001	0.587 $\pm$ 0.005	0.2

datasets, and TabPFN-Unsupervised requires several minutes. AnoLLM and TabPFN-Unsupervised therefore demand orders of magnitude more computation without a corresponding gain in detection quality. In answer to RQ1, the TFMs do not outperform established baseline detectors in the unsupervised setting.

C. Experiment 2: Representation Variants and Enhanced Baselines

Experiment 2 addresses RQ2. It takes the baseline detectors from Experiment 1 and replaces the text free cleaned input by the representation variants of Section III-B, following the pipeline in Figure 1. The grid search is repeated for every representation, so that no detector inherits a configuration that was selected on a different input space. The `cleaned` variant from Experiment 1 serves as the reference. The comparison covers the two external embedding families, the sentence transformer and FastText, and the semi-supervised enhanced variants.

On the text-rich Fake Job Postings dataset the FastText variant reduced to 30 components lies at or above the cleaned reference for all four detectors, and the gain exceeds the spread across the seeds for Isolation Forest, which rises from 0.084 to 0.115. For the remaining detectors the point estimate is higher than the cleaned reference but the difference stays within one standard deviation, so the consistent direction across all four detectors, rather than the size of the individual gain, is what carries the observation. The standard deviations of all values are reported in Table XII and Table XIII in the appendix. On the more numeric Airbnb Paris dataset the same variant stays below the cleaned reference for every detector. The sentence transformer variants exceed the cleaned reference only for Isolation Forest on Fake Job Postings and stay below it in all other cases. A reduction to 30 components is stronger than a reduction to 100 components in 15 of the 16 comparisons. Among the unsupervised external embeddings FastText is therefore the strongest, which matches the finding

of [9] that FastText performs best most often. The benefit is confined to the dataset with the higher share of free text, so no external embedding strategy improves the detection across both datasets. Table VII reports the corresponding AUROC values and shows the same dependence on the dataset, with the standard deviations again reported in the appendix. The result holds for two structurally different embedding procedures, which indicates that the absence of a consistent benefit does not stem from the specific embedding model but from the unsupervised transfer of the free text into the input space of the detectors. This is the point at which the finding of this work departs from the supervised result of Mráz et al., where such embeddings improved performance on the benchmark as a whole.

The enhanced variants behave differently on the two datasets. On Fake Job Postings the best enhanced variant of each detector reaches the highest values in Table VI and exceeds both the cleaned reference and the external embedding variants. On Airbnb Paris only the autoencoder improves, while ECOD, Isolation Forest and LODA stay below the cleaned reference. This distance must not be read as a competing method, since the enhanced variants carry label information and are semi-supervised, as Section III-B states. Their value is diagnostic. On the text-rich dataset they show that the detectors do draw a benefit from an enriched numeric representation, provided that the representation makes the information accessible in a form the detectors can use. The problem therefore lies not in the use of representations as such but in their origin. On Airbnb Paris even the label bearing representation does not improve the detection for three of the four detectors, which indicates that the limit on that dataset is not only the absence of a label. This reading is supported by Experiment 3, where the supervised classifiers reach a far lower AUPRC on Airbnb Paris than on Fake Job Postings, as Table VIII shows. The AUROC on Airbnb Paris is high, so the constructed label is well determined by the tabular attributes as far as a ranking is concerned. What the label does not provide is a geometric separation of the anomalous rows in the input space, which is what an unsupervised detector requires. A label bearing representation therefore encodes a distinction that the detectors cannot exploit once the label is removed. The weak enhanced result on this dataset and the low supervised ceiling in Experiment 3 thus share a single cause in the label rather than in the representation. The combination `enhanced_semantic_pca30` does not consistently exceed the better of its two sources, which suggests that the two representation sources contribute largely redundant rather than complementary information. In answer to RQ2, no representation strategy improves the detection across both datasets, and the one strategy that helps on the text-rich dataset, namely FastText reduced to 30 components, does not transfer to the more numeric one.

D. Experiment 3: Foundation Models for Binary Classification

Experiment 3 addresses RQ3 and serves as a control for the previous findings. The weak unsupervised performance

TABLE VI

EXPERIMENT 2. AUPRC PER DETECTOR AND REPRESENTATION VARIANT. THE `CLEANED` COLUMN IS THE TEXT FREE REFERENCE FROM EXPERIMENT 1. THE ENHANCED COLUMNS ARE SEMI-SUPERVISED AND ARE REPORTED AS A DIAGNOSTIC UPPER BOUND. THE BEST UNSUPERVISED VALUE AND THE BEST SEMI-SUPERVISED VALUE PER DETECTOR AND DATASET ARE SET IN BOLD.

	Unsupervised					semi-supervised		
	cleaned	sem. p30	sem. p100	ft. p30	ft. p100	enh.	enh. p30	enh. sem. p30
<i>Fake Job Postings</i>								
Autoencoder	0.074	0.067	0.048	0.078	0.073	0.120	0.083	0.082
ECOD	0.076	0.054	0.043	0.085	0.068	0.120	0.110	0.074
Isolation Forest	0.084	0.089	0.052	0.115	0.081	0.119	0.132	0.118
LODA	0.055	0.052	0.045	0.065	0.068	0.103	0.113	0.075
<i>Airbnb Paris</i>								
Autoencoder	0.068	0.055	0.053	0.059	0.045	0.075	0.071	0.076
ECOD	0.100	0.078	0.058	0.075	0.051	0.039	0.076	0.085
Isolation Forest	0.094	0.080	0.076	0.081	0.063	0.049	0.074	0.088
LODA	0.084	0.056	0.051	0.081	0.077	0.040	0.065	0.051

TABLE VII

EXPERIMENT 2. AUROC PER DETECTOR AND REPRESENTATION VARIANT. THE `CLEANED` COLUMN IS THE TEXT FREE REFERENCE FROM EXPERIMENT 1. THE ENHANCED COLUMNS ARE SEMI-SUPERVISED AND ARE REPORTED AS A DIAGNOSTIC UPPER BOUND. THE BEST UNSUPERVISED VALUE AND THE BEST SEMI-SUPERVISED VALUE PER DETECTOR AND DATASET ARE SET IN BOLD.

	Unsupervised					semi-supervised		
	cleaned	sem. p30	sem. p100	ft. p30	ft. p100	enh.	enh. p30	enh. sem. p30
<i>Fake Job Postings</i>								
Autoencoder	0.615	0.602	0.517	0.627	0.620	0.720	0.617	0.667
ECOD	0.626	0.544	0.475	0.608	0.611	0.761	0.680	0.639
Isolation Forest	0.619	0.643	0.525	0.682	0.646	0.760	0.763	0.719
LODA	0.504	0.551	0.477	0.593	0.615	0.722	0.736	0.625
<i>Airbnb Paris</i>								
Autoencoder	0.645	0.567	0.546	0.569	0.558	0.711	0.686	0.650
ECOD	0.741	0.690	0.606	0.689	0.614	0.523	0.716	0.733
Isolation Forest	0.721	0.692	0.669	0.690	0.662	0.567	0.704	0.725
LODA	0.655	0.574	0.559	0.657	0.646	0.540	0.634	0.558

in Experiment 1 admits two explanations. Either the representations of the TFMs are of insufficient quality, or the representations are informative but the unsupervised procedure fails to exploit them. To separate these explanations the same models are applied to the same data as a supervised binary classification task, in which a label guides the model. TabPFN and ConTextTab are evaluated with two compositions of the in-context training set, both drawn at an equal size of 1500 rows so that the comparison isolates the effect of the class balance from the effect of the context size. The natural variant draws 1500 rows from the training part in its natural distribution, which yields about 73 outliers on Fake Job Postings and about 59 on Airbnb Paris. The balanced variant, denoted balanced 1:4, draws 300 outliers and 1200 inliers from the same training part. The term balanced refers to the composition of the in-context training set alone. The test part remains unchanged in its natural distribution in both variants, so that the reported values stay comparable with each other and with Experiment 1 and Experiment 2.

Table VIII reports AUPRC, AUROC and runtime for both compositions of the in-context training set. In contrast to the unsupervised setting both models reach higher values. On Fake Job Postings ConTextTab exceeds an AUROC of

0.96 in both compositions and an AUPRC of 0.90 in the balanced one, and on Airbnb Paris both models exceed an AUROC of 0.85. For TabPFN, the same representations that were of little use in the unsupervised setting are therefore highly informative once a label directs the model towards the relevant part of the representation. This locates the obstacle. The weak unsupervised performance does not originate from insufficient representation quality but from the adaptation to the unsupervised procedure. In answer to RQ3, the representations become effective under supervision, which confirms that the bottleneck lies in the unsupervised transfer rather than in the representations themselves. ConTextTab is the stronger classifier on the text-rich dataset, while the two models are close on the more numeric dataset.

The balanced composition outperforms the natural one for both models and on both datasets, and the gap is widest on the text-rich dataset. Since the size of the in-context training set is held constant, the number of anomalous examples in that set limits the classifiers more than its absolute size does. This is consistent with the in-context nature of both models, since only the examples in the context are available to the model at inference.

TABLE VIII

EXPERIMENT 3. SUPERVISED BINARY CLASSIFICATION AS A CONTROL, AT AN EQUAL IN-CONTEXT SIZE OF 1500 ROWS FOR BOTH COMPOSITIONS. AUPRC, AUROC (WITH STANDARD DEVIATIONS OVER THREE SEEDS) AND RUNTIME IN SECONDS. BEST AUPRC AND AUROC PER DATASET IN BOLD.

Data	Model	AUPRC $\pm$ SD	AUROC $\pm$ SD	Time
F. Jobs	ConTextTab bal. 1:4	0.911 $\pm$ 0.012	0.988 $\pm$ 0.003	21.9
	ConTextTab natural	0.781 $\pm$ 0.043	0.970 $\pm$ 0.008	22.4
	TabPFN bal. 1:4	0.699 $\pm$ 0.016	0.946 $\pm$ 0.006	4.1
	TabPFN natural	0.562 $\pm$ 0.012	0.919 $\pm$ 0.005	4.3
Airbnb	ConTextTab bal. 1:4	0.231 $\pm$ 0.022	0.889 $\pm$ 0.007	22.4
	TabPFN bal. 1:4	0.227 $\pm$ 0.020	0.889 $\pm$ 0.011	5.2
	TabPFN natural	0.185 $\pm$ 0.015	0.861 $\pm$ 0.005	5.2
	ConTextTab natural	0.183 $\pm$ 0.015	0.854 $\pm$ 0.006	22.5

V. DISCUSSION

The four sets of results combine into a single picture. The preliminary analysis in Section IV-A shows that the free text columns carry information that is relevant for the prediction, more strongly so on the text-rich dataset. Experiment 1 shows that the TFMs reach no consistent advantage over tuned baseline detectors in the unsupervised setting. Their runtimes differ widely, since AnoLLM and TabPFN-Unsupervised demand orders of magnitude more computation than the baseline detectors, whereas FoMo-0D is the fastest model of the comparison. Experiment 2 shows that neither external embedding family makes the free text usable across both datasets. FastText improves the detection on the text-rich dataset, whereas no variant improves it on the more numeric one, and the finding holds across two structurally different embedding procedures. Experiment 3 shows that the same models and representations become highly informative once a label directs them.

Read together, these results locate the obstacle. The free text is not irrelevant, and the representations of the TFMs are not of insufficient quality, since both become effective under supervision. The two datasets therefore fail for different reasons. On the text rich dataset the information is present and only its transfer into an unsupervised procedure fails. On the more numeric dataset the constructed label separates the classes in a ranking but not in the geometry of the input space, so no representation can make it accessible to an unsupervised detector there. The enhanced variants make this precise. They improve the detection substantially on the text-rich dataset, but they draw their advantage from label information and are therefore semi-supervised. Their role is diagnostic rather than practical. They demonstrate that the detectors can exploit an enriched numeric representation, provided that the representation makes the information accessible. The unsupervised text embeddings achieve this only on the text rich dataset.

This result stands in a clear relation to the study of Mráz et al. [9]. In the supervised case they observe that text embeddings improve predictive performance, whereas in the unsupervised case studied here the same kind of embeddings does not. The two findings are consistent once the role of

the label is taken into account. The label is what turns the additional text information into a usable signal, and its absence, not the text or the embedding, is what removes the benefit.

A behaviour worth noting concerns AnoLLM. It reaches the best result on the text-rich Fake Job Postings dataset and falls behind on the more numeric Airbnb dataset. This matches the design of LLM based approaches, whose strength lies in the processing of free text, and it is consistent both with the different shares of text in the two datasets and with the privileged input that AnoLLM receives in Experiment 1. The low absolute AUPRC values across all models further reflect the difficulty of unsupervised anomaly detection under strong class imbalance. The higher AUROC values indicate that the models establish a coarse ranking between normal and anomalous points, while the precise identification of the rare anomalies remains hard for every approach.

For unsupervised anomaly detection on heterogeneous tabular data of this kind, established baseline detectors remain the more appropriate choice for now. None of the TFMs reaches a higher detection quality, and the two that are expensive to run add a substantial computational cost without a corresponding gain.

A. Limitations

Several limitations qualify these conclusions. On the data side, the evaluation rests on two datasets from two domains, so statements about the generalisability of the findings, in particular the different effect of the PCA on the two embedding families, are empirically confined to these two cases. In addition, the Airbnb dataset has no native anomaly label. The label was constructed by treating a rating of 5.0 as normal and a rating of at most 3.0 as anomalous, which is a deliberate but arbitrary operationalisation, and no sensitivity analysis over alternative thresholds was conducted.

On the technical side, the available GPU memory prevented the evaluation of the semantic and FastText variants without a preceding PCA, so the findings on the sensitivity to the PCA are limited to the reduced variants. For the same reason the context of TabPFN-Unsupervised was capped at 3000 rows and drawn at random from the training part rather than in a stratified manner. The evaluation covers the full test part, so the limitation concerns the information available to the model and not the basis of the measurement. Since the anomalies enter the context at their natural rate, a random draw of this size contains few of them, which is a plausible explanation for the ranking close to chance on Fake Job Postings.

On the methodological side, the following four apply. The model selection reflects the current landscape, in which many TFMs are either not designed for anomaly detection or cannot process text natively, so the comparison is restricted to the few models that fit the task. The asymmetry of the evaluation protocol, in which the baseline detectors were tuned by grid search while the TFMs were run in their default configurations, was chosen to avoid unfavourable standard hyperparameters and must be kept in mind when the relative

results are read. A further asymmetry concerns the enhanced representations. Their TabPFN embeddings are fitted on the label by construction, so these variants are semi-supervised and are not comparable with the unsupervised variants. They are reported as a diagnostic upper bound rather than as a competing method, and the values they reach are not attainable in a setting without labels. Finally, the attribution analysis was performed on a classification model rather than on a detector, since SHAP values require a model of sufficient quality. It therefore explains the prediction of that model, which is why it is used only to establish the relevance of the free text and not as evidence about the detectors themselves. The attribution ranking is reported as the mean and standard deviation over the three seeds, so the reported spread understates the full uncertainty of the ranking, since it does not account for the choice of background samples and explained rows.

A further point concerns the runtime budget. The number of optimisation steps of AnoLLM was reduced to 1 000 to keep the runtime manageable, so the reported values of AnoLLM are a lower bound on the performance that the model reaches with its default budget.

The balance controlled comparison in Experiment 3 is conducted at a single context size of 1500 rows. This size was chosen because the balanced 1:4 composition is bounded by the number of outliers in the training part, which is smallest on Airbnb Paris at about 360 and caps a 1:4 context at about 1800 rows. The interaction between context size and class balance across a wider range of sizes was not examined and is left to future work.

VI. CONCLUSION AND FUTURE WORK

This work investigated the potential of TFMs for multivariate anomaly detection on tabular data and the extent to which the semantic information of free text columns can be made usable for this task. Along three research questions, a preliminary analysis and three experiments were conducted on two datasets with different shares of free text.

The answers combine into a single conclusion. In answer to RQ1, the TFMs reach no consistent advantage over established baseline detectors in the unsupervised setting. Where they are substantially more expensive to run, as for AnoLLM and TabPFN-Unsupervised, this cost is not offset by a higher detection quality. In answer to RQ2, none of the studied representation strategies makes the free text usable across both datasets. FastText reduced to 30 components improves the detection on the text-rich dataset, but the benefit does not transfer to the more numeric one, and this holds for two structurally different embedding procedures. The preliminary analysis shows that this is not caused by irrelevant text, since the free text columns carry measurable and, on the text-rich dataset, dominant information. In answer to RQ3, TabPFN reaches high values under supervision after performing weakly in the unsupervised setting, which shows that its representations are informative and that the weak unsupervised performance stems from the adaptation to the unsupervised procedure rather than from the quality of the representations. ConTextTab, which is

not evaluated as an unsupervised detector, reaches high values as well and thereby confirms that the representations of a TFM can be highly informative once a label is available.

The contribution of this work therefore lies less in a new detection method than in a systematic empirical classification. It compares TFMs and baseline detectors on a common data basis, isolates the influence of different representation forms and makes the tension between statistical exactness and semantic density visible for the unsupervised case. In doing so it transfers the question studied by Mráz et al. [9] from the supervised to the unsupervised setting and shows that the benefit of text embeddings does not carry over. The central finding is that the obstacle lies not in the representation quality of the TFMs but in the transfer of these representations into an unsupervised detection procedure that preserves the semantic information.

Three directions for future work follow directly from this finding. The first concerns the design of the models. Since the representations are informative but their transfer fails, future TFMs for this task should not be derived from classification models but should be trained with objectives and architectures that target unsupervised anomaly detection from the outset. A central building block is the extraction of semantic representations without a label, since the enhanced variants of Experiment 2 gained their advantage precisely from the label information they carried.

The second concerns the data. As stated in the motivation, openly available multivariate datasets that combine numeric and textual attributes with a native anomaly label are scarce [9]. This scarcity confined the present evaluation to two datasets and required the construction of a label for the Airbnb dataset. Progress in the field therefore depends on more open datasets with this combination of attributes and on shared benchmarks that allow a systematic and comparable evaluation across several domains.

The third concerns the integration of the free text into the training. The preliminary analysis showed that the free text carries relevant information, while a subsequent embedding does not make this information accessible to the detectors. A promising route is therefore to account for the text semantics already during training rather than to append it afterwards. ConTextTab, which is trained on real tables and converts textual cells into vectors while preserving the column semantics, offers a starting point for such an approach.

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USE OF AI TOOLS

During the preparation of this work, Claude (by Anthropic) was used for language translation and text revision of the manuscript, and Claude Code (by Anthropic) was used for code implementation support during the execution of the experiments. All experimental designs, data analyses, result interpretations, and final decisions remain entirely the author's

own, and the author takes full responsibility for the submitted work.

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APPENDIX

TABLE IX

SHARE OF VARIANCE RETAINED BY THE PCA REDUCTION OF THE TWO EXTERNAL TEXT EMBEDDING VARIANTS. THE VALUES ARE AVERAGED OVER THE THREE SEEDS, SINCE THE PCA IS FITTED ON THE TRAINING ROWS OF EACH SPLIT.

Embedding	Dataset	30 comp.	100 comp.
Sentence transformer	Fake Job Postings	47 %	64 %
Sentence transformer	Airbnb Paris	57 %	72 %
FastText	Fake Job Postings	68 %	87 %
FastText	Airbnb Paris	77 %	93 %

TABLE X
COMPARISON OF THE STUDIED FMS.

Property	TabPFN-3	TabPFN Unsup.	FoMo-0D	ConTextTab	AnoLLM
Primary use case	Classification and regression (supervised)	Outlier detection and representation extraction (unsupervised)	Outlier detection (unsupervised)	Classification and regression	Anomaly detection with textual semantics
Paradigm	PFN, synthetic priors, ICL	PFN, synthetic priors, same base as TabPFN-3	PFN, synthetic priors, zero-shot inference	Hybrid, trained on real tables, table native ICL	LLM based, row serialisation into text
Max. features	$\approx 2,000$	$\approx 2,000$	$\approx 100-500$	$\approx 100-200$	$\approx 100-200$
Max. rows	$\approx 50,000$	limited by context size	$\approx 5,000-10,000$	token based context window	single row inference
Role in this work	Supervised classification control	Detector and source of the enhanced representations	Direct point of comparison for RQ1	Attribution basis and classification control	Detector, counterpart of the table native models

TABLE XI
GRID SEARCH SPACE OF THE BASELINE ANOMALY DETECTORS.

Model	Search space
iForest	$n_estimators \in \{100, 200\}, max_features \in \{0.5, 1.0\}$
LODA	$n_bins \in \{10, 20\}, n_random_cuts \in \{100, 200\}$
ECOD	parameter free
AutoEncoder	$hidden_neuron_list \in \{[64, 32], [32, 16]\}, epoch_num \in \{20, 50\}$

TABLE XII

EXTENDED VERSION OF TABLE VI (AUPRC, EXPERIMENT 2) WITH MEAN AND STANDARD DEVIATION OVER THREE SEEDS. THE CLEANED COLUMN REPEATS THE REFERENCE VALUES FROM TABLE V. THE BEST UNSUPERVISED VALUE AND THE BEST SEMI-SUPERVISED VALUE PER DETECTOR AND DATASET ARE SET IN BOLD.

	Unsupervised					semi-supervised		
	cleaned	sem. p30	sem. p100	ft. p30	ft. p100	enh.	enh. p30	enh. sem. p30
<i>Fake Job Postings</i>								
Autoencoder	0.074 $\pm$ 0.002	0.067 $\pm$ 0.008	0.048 $\pm$ 0.003	0.078 $\pm$ 0.007	0.073 $\pm$ 0.009	0.120 $\pm$ 0.011	0.083 $\pm$ 0.006	0.082 $\pm$ 0.009
ECOD	0.076 $\pm$ 0.004	0.054 $\pm$ 0.001	0.043 $\pm$ 0.000	0.085 $\pm$ 0.007	0.068 $\pm$ 0.009	0.120 $\pm$ 0.014	0.110 $\pm$ 0.042	0.074 $\pm$ 0.008
Isolation Forest	0.084 $\pm$ 0.010	0.089 $\pm$ 0.005	0.052 $\pm$ 0.006	0.115 $\pm$ 0.013	0.081 $\pm$ 0.015	0.119 $\pm$ 0.019	0.132 $\pm$ 0.009	0.118 $\pm$ 0.018
LODA	0.055 $\pm$ 0.016	0.052 $\pm$ 0.005	0.045 $\pm$ 0.005	0.065 $\pm$ 0.011	0.068 $\pm$ 0.012	0.103 $\pm$ 0.016	0.113 $\pm$ 0.033	0.075 $\pm$ 0.023
<i>Airbnb Paris</i>								
Autoencoder	0.068 $\pm$ 0.009	0.055 $\pm$ 0.001	0.053 $\pm$ 0.005	0.059 $\pm$ 0.012	0.045 $\pm$ 0.001	0.075 $\pm$ 0.007	0.071 $\pm$ 0.006	0.076 $\pm$ 0.014
ECOD	0.100 $\pm$ 0.011	0.078 $\pm$ 0.006	0.058 $\pm$ 0.003	0.075 $\pm$ 0.002	0.051 $\pm$ 0.002	0.039 $\pm$ 0.004	0.076 $\pm$ 0.007	0.085 $\pm$ 0.005
Isolation Forest	0.094 $\pm$ 0.019	0.080 $\pm$ 0.010	0.076 $\pm$ 0.009	0.081 $\pm$ 0.012	0.063 $\pm$ 0.004	0.049 $\pm$ 0.004	0.074 $\pm$ 0.012	0.088 $\pm$ 0.007
LODA	0.084 $\pm$ 0.021	0.056 $\pm$ 0.008	0.051 $\pm$ 0.003	0.081 $\pm$ 0.012	0.077 $\pm$ 0.015	0.040 $\pm$ 0.002	0.065 $\pm$ 0.017	0.051 $\pm$ 0.003

TABLE XIII

EXTENDED VERSION OF TABLE VII (AUROC, EXPERIMENT 2) WITH MEAN AND STANDARD DEVIATION OVER THREE SEEDS. THE `CLEANED` COLUMN REPEATS THE REFERENCE VALUES FROM TABLE V. THE BEST UNSUPERVISED VALUE AND THE BEST SEMI-SUPERVISED VALUE PER DETECTOR AND DATASET ARE SET IN BOLD.

	Unsupervised					semi-supervised		
	cleaned	sem. p30	sem. p100	ft. p30	ft. p100	enh.	enh. p30	enh. sem. p30
<i>Fake Job Postings</i>								
Autoencoder	0.615±0.011	0.602±0.018	0.517±0.022	0.627 ±0.020	0.620±0.027	0.720 ±0.028	0.617±0.044	0.667±0.020
ECOD	0.626 ±0.009	0.544±0.008	0.475±0.004	0.608±0.023	0.611±0.033	0.761 ±0.024	0.680±0.046	0.639±0.035
Isolation Forest	0.619±0.015	0.643±0.002	0.525±0.033	0.682 ±0.016	0.646±0.036	0.760±0.025	0.763 ±0.007	0.719±0.028
LODA	0.504±0.069	0.551±0.040	0.477±0.032	0.593±0.046	0.615 ±0.037	0.722±0.047	0.736 ±0.046	0.625±0.076
<i>Airbnb Paris</i>								
Autoencoder	0.645 ±0.017	0.567±0.009	0.546±0.007	0.569±0.013	0.558±0.002	0.711 ±0.020	0.686±0.021	0.650±0.017
ECOD	0.741 ±0.014	0.690±0.004	0.606±0.002	0.689±0.004	0.614±0.006	0.523±0.028	0.716±0.014	0.733 ±0.006
Isolation Forest	0.721 ±0.013	0.692±0.007	0.669±0.006	0.690±0.017	0.662±0.010	0.567±0.009	0.704±0.007	0.725 ±0.007
LODA	0.655±0.021	0.574±0.032	0.559±0.008	0.657 ±0.050	0.646±0.024	0.540±0.008	0.634 ±0.078	0.558±0.015

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Titel der Arbeit: Evaluation von Embeddings aus Foundation Modellen zur Ausreißererkennung aus multivariaten tabellarischen Daten

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